

from Earth-based telescopes. The proliferation of these small satellites, coupled with existing space debris, raises concerns about a potential Kessler effect, where the density of objects in space reaches a point where collisions generate further debris, making space exploration and utilisation more difficult.

In this evolving landscape, the threat of cyber-attacks on space assets cannot be ignored. Cyber capabilities can cause significant damage to space infrastructure at relatively low costs. These attacks can target ground control systems, intercept satellite signals, or compromise various electronic components. Attribution of such cyber activities can be challenging, allowing both state and non-state actors to potentially access and exploit vulnerabilities in space systems. Ensuring robust cybersecurity measures, integrating security standards during the design process, and fostering international collaboration are vital to safeguard space assets from cyber threats.

Addressing the safety and security of space assets requires a multi-dimensional approach. Technological advancements and innovations play a crucial role in enhancing the resilience of space systems. Simultaneously, legal frameworks need to be developed to establish accountability and liability for actions that pose risks to space systems, including collisions, failures, and cyber interferences. International organisations such as the International Telecommunication Union (ITU) are involved in defining regulations and standards for space activities, including the allocation of orbits and frequencies to prevent interference between different actors

Legal framework for cyber-secure large constellations

In an ambitious quest to deliver internet connectivity to every corner

of the globe, grand constellations of satellites endeavour to revolutionise global access with lightning-fast speeds and immense capacity. However, this bold endeavour brings with it a lurking peril—the looming spectre of cyber assaults targeting not only the celestial platforms themselves but also the precious assets they harbour.

Imagine a celestial ballet unfolding high above as a myriad of interconnected satellites form an intricate web to bridge the digital divide. These celestial pioneers, traversing the heavens, strive to create a global network that knows no boundaries. Their noble mission promises a world where connectivity transcends geographical limitations, where the remotest regions are embraced by the digital fabric of modern society.

Yet, in this breathtaking celestial symphony, a sinister undertone emerges. The very attributes that make these constellations a technological marvel also render them vulnerable to the insidious reach of cyber adversaries. Like shadows lurking in the vast expanse of space, cyber attackers lie in wait, poised to exploit any weaknesses that may permeate the cosmic infrastructure.

The stakes are high, for these interconnected constellations possess the power to reshape the digital landscape as we know it. Their potential to deliver low-latency, high-capacity internet services holds the key to unlocking unprecedented opportunities for individuals, businesses, and societies. However, this promise of boundless connectivity comes hand in hand with the grave risk of cyber onslaughts.

The celestial platforms, those beacons of connectivity that dot the night sky, stand as prime targets for malicious actors seeking to disrupt the delicate balance of the digi-

tal realm. Their intricate network architecture, the intricate interplay of signals and data coursing through the cosmic pathways, becomes an enticing playground for cyber assailants. With a single stroke of malevolence, they could sow chaos, severing the lifelines that connect individuals and communities, and cast a dark cloud over the promise of a connected world.

But it is not only the platforms themselves that are at risk. The precious assets carried by these celestial envoys, the satellites that traverse the celestial expanse, are also in the crosshairs of cyber adversaries. These technological marvels, teeming with state-of-the-art systems and sensitive information, become tantalising targets for those with malicious intent. Their vulnerability in the vastness of space leaves them exposed to the clandestine manoeuvres of hackers, who seek to exploit any vulnerability, jeopardising the very fabric of our interconnected world.

As humanity reaches for the stars in its quest for universal connectivity, we must not lose sight of the perils that lurk in the celestial abyss. Cyber defences must stand as stalwart guardians, protecting the celestial pioneers and their precious cargo from the looming shadows of the digital realm. Only by fortifying the cosmic infrastructure, and by fortifying our resolve to combat cyber threats, can we ensure that the celestial ballet continues its graceful performance, illuminating the world with the transformative power of connectivity, unmarred by the terrors of the virtual underworld.

Definition and legality of the large constellations of satellites

Embark on a captivating journey through the cosmos, where an intricate web of artificial satellites,